

A Reputation-Based Incentive and Punishment Mechanism for Campus WiFi Sharing Based on Evolutionary Game Theory

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Abstract—To address the free-riding and insufficient contribution issues in campus WiFi sharing, this paper proposes a reputation-threshold-based incentive and punishment mechanism founded on evolutionary game theory. The proposed model integrates continuous and discrete replicator dynamics with a finite-population Moran process to analyze the evolutionary behavior and fixation probability of sharing and free-riding strategies. In addition, by leveraging dynamic authorization in the Remote Authentication Dial-In User Service (RADIUS) framework, the reputation-based incentive and punishment mechanism is mapped to deployable control actions, including session rate limiting, priority adjustment, and re-authentication enforcement. Furthermore, this paper provides a reproducible simulation workflow, parameter settings, and an evaluation metric system.

Index Terms—campus network, WiFi sharing, evolutionary game theory, Moran process, reputation mechanism, incentive and punishment mechanism, RADIUS CoA.

I. INTRODUCTION

CAMPUS WiFi access is commonly managed through the Authentication, Authorization, and Accounting (AAA) framework over IEEE 802.11 networks. In practice, EAP supports user authentication at the link layer, while RADIUS carries authorization and session-control parameters after authentication. This architecture provides a feasible engineering path for translating incentive mechanisms into deployable network policies [1]–[3].

In campus WiFi sharing, users may consume network resources while avoiding contribution, leading to the classical free-rider problem. Among existing incentive approaches, reputation-based mechanisms are particularly suitable for campus networks because they are identity-verifiable, administratively manageable, and relatively low-cost to deploy. However, existing studies often lack clear implementation paths and reproducible evaluation settings [4].

Meanwhile, modern AAA systems can enforce differentiated bandwidth and access policies and support dynamic authorization updates through Change of Authorization (CoA). Therefore, reputation scores can be mapped to practical session-level actions, such as bandwidth promotion, rate limiting, priority adjustment, and forced re-authentication [5], [6].

Game-theoretic resource management has also been applied to a variety of wireless communication scenarios. Evolutionary

games have been employed to optimize IRS reflection coefficients by allowing users or control entities to adapt strategies according to payoff differences [11]. Deep reinforcement learning combined with evolutionary games has been used for dynamic spectrum access under changing channel conditions [12]. Auction mechanisms have also been introduced for dynamic user allocation in heterogeneous networks [13]. These studies demonstrate the applicability of adaptive game-theoretic control to wireless resource management, but they do not directly address reputation-driven campus WiFi sharing.

Based on this observation, this paper proposes a reputation-threshold incentive and punishment framework for campus WiFi sharing under evolutionary game theory, combining replicator dynamics, the Moran process, and a reproducible simulation design for practical evaluation.

II. EVOLUTIONARY GAME-THEORETIC FOUNDATIONS

To characterize the long-term strategic adaptation of campus network users between sharing and free-riding, this paper adopts three evolutionary models: continuous-time replicator dynamics, discrete-time replicator difference equations, and the finite-population Moran process. Let the strategy set be $S = \{S, D\}$, where S denotes sharing and D denotes defection. Let $x(t) \in [0, 1]$ denote the proportion of users adopting the sharing strategy at time t , and let $r_i(t) \in [0, 1]$ denote the reputation score of user i [7]. The use of evolutionary adaptation is consistent with wireless optimization studies in which agents iteratively adjust resource strategies according to payoff differences [11].

For a finite strategy set, the continuous replicator dynamics can be written as

$$\dot{x}_i = x_i (\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (1)$$

where

$$\bar{\pi}(\mathbf{x}) = \sum_j x_j \pi_j(\mathbf{x}). \quad (2)$$

This formulation implies that a strategy increases in population share when its payoff exceeds the population average, and decreases otherwise. Therefore, replicator dynamics provide a natural description of strategy imitation and population evolution under repeated interaction [7].

For the two-strategy case $S = \{S, D\}$, the dynamics reduce to

$$\dot{x} = x(1-x) (\pi_S(x) - \pi_D(x)). \quad (3)$$

The equilibrium points satisfy either $x \in \{0, 1\}$ or $\pi_S(x) = \pi_D(x)$. The boundary equilibria correspond to the pure states

This work was completed as a student research project. Project implementation is available on GitHub: https://github.com/cob333/Tele_Wifi.

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of full sharing and full defection, whereas interior fixed points represent mixed coexistence states. Their local stability can be determined from the derivative of the payoff-difference function [7].

To align the model with slot-based reputation updating in practical systems, the continuous dynamics can be discretized by the Euler method as

$$x_{t+1} = x_t + \eta x_t (1 - x_t) (\pi_S(x_t) - \pi_D(x_t)), \quad (4)$$

where $\eta > 0$ is the update step size. To ensure physical interpretability, the updated proportion should remain within $[0, 1]$, which can be enforced by truncation or normalization when necessary [7].

To further capture stochastic perturbations and finite-size effects in a campus population of limited scale, this paper also considers the Moran process. Let $i \in \{0, \dots, N\}$ denote the number of sharing users in a population of size N . Under the standard birth-death update rule, one individual is selected for reproduction according to fitness and then replaces another randomly selected individual. The corresponding transition probabilities are

$$T_i^+ = \frac{if_S(i)}{if_S(i) + (N-i)f_D(i)} \frac{N-i}{N}, \quad (5)$$

and

$$T_i^- = \frac{(N-i)f_D(i)}{if_S(i) + (N-i)f_D(i)} \frac{i}{N}, \quad (6)$$

where f_S and f_D are the fitness values induced by the payoff functions [8]. Compared with deterministic replicator dynamics, the Moran process is more suitable for analyzing absorption behavior, fixation outcomes, and stochastic evolutionary trajectories in small and medium-sized populations.

In a general birth-death chain, the fixation probability of a single mutant is given by

$$\rho_1 = \left(1 + \sum_{k=1}^{N-1} \prod_{j=1}^k \frac{T_j^-}{T_j^+} \right)^{-1}. \quad (7)$$

This quantity measures the probability that a mutant strategy eventually takes over the whole population. Together with the concept of an evolutionarily stable strategy (ESS), it provides a complementary perspective to the local asymptotic stability of replicator dynamics. In particular, if an interior fixed point $x^* \in (0, 1)$ is attracting, sharing and defection may coexist stably within a certain parameter range; if the sharing strategy exhibits a higher fixation probability or satisfies the ESS condition, the proposed mechanism is more likely to suppress free-riding in long-term evolution [7], [9].

III. REPUTATION-BASED INCENTIVE AND PUNISHMENT MECHANISM

The proposed system consists of a user set $\mathcal{U} = \{1, \dots, N\}$, an access-point set $\mathcal{A} = \{1, \dots, M\}$, and a platform-side control domain including AAA, accounting, and reputation-management services. At each time slot, each user selects either sharing (S) or defection (D), and updates its strategy in the next slot according to the realized utility and reputation

evolution [3]. Dynamic association and allocation among users and access resources are also central concerns in auction-based heterogeneous-network models [13]; in the present mechanism, reputation replaces monetary bids as the primary control signal.

Let $Q_i(t)$ denote the composite quality of service perceived by user i at time t , which reflects network performance factors such as throughput and delay, and let $c_i(t)$ denote the cost of sharing, including energy consumption, bandwidth occupation, and potential security risk. The utility of user i is defined as

$$u_i(t) = vQ_i(t) - c_i(t)\mathbf{1}_{\{a_i(t)=S\}} + R(r_i(t))\mathbf{1}_{\{a_i(t)=S\}} - P(r_i(t))\mathbf{1}_{\{a_i(t)=D\}}, \quad (8)$$

where $a_i(t) \in \{S, D\}$ is the selected strategy, $r_i(t) \in [0, 1]$ is the reputation score, and v is the service-valuation coefficient [9]. Based on the induced expected payoffs π_S and π_D , the payoff difference

$$\Delta\pi = \pi_S - \pi_D \quad (9)$$

serves as the key quantity driving both replicator-based updates and finite-population evolutionary transitions.

To jointly account for historical contribution and resource consumption, a contribution-to-consumption signal is introduced as

$$\phi_i(t) = \frac{\text{share}_i(t)}{\text{use}_i(t) + \epsilon}, \quad (10)$$

where $\epsilon > 0$ is a small constant for numerical stability. The reputation score is then updated by

$$r_i(t+1) = \text{clip}((1-\lambda)r_i(t) + \lambda\sigma(\phi_i(t) - \kappa), 0, 1), \quad (11)$$

where $\lambda \in (0, 1]$ controls the tradeoff between historical reputation and current observation, κ is the contribution-ratio baseline, and $\sigma(\cdot)$ can be implemented as a sigmoid or piecewise-linear mapping. This smoothed update suppresses excessive fluctuations caused by noisy short-term observations and is convenient for robustness evaluation under imperfect monitoring.

A threshold-based reward and punishment mechanism is adopted to make reputation an effective control variable rather than a descriptive metric only. Let $\theta \in [0, 1]$ denote the reputation threshold. The reward and punishment functions are defined as

$$R(r) = \alpha[r - \theta]^+, \quad P(r) = \gamma[\theta - r]^+, \quad (12)$$

where $[z]^+ = \max(z, 0)$, and $\alpha, \gamma \geq 0$ denote the reward and punishment intensities, respectively [4]. When $r \geq \theta$, the user receives positive incentives; otherwise, restrictive control is imposed. Through $R(r)$ and $P(r)$, the parameters θ , α , and γ directly reshape the payoff difference $\Delta\pi$, thereby affecting the evolutionary direction and equilibrium structure of the system.

To ensure practical deployability, three types of constraints are considered. First, the total reward expenditure in each slot is bounded by a budget constraint,

$$\sum_i R(r_i(t)) \leq B_{\max}. \quad (13)$$

Second, imperfect behavior observation is modeled through the detection probability p_d and false-punishment probability p_f , which are explicitly incorporated in simulation. Third, each access point is subject to a capacity constraint,

$$\sum_{i \in \mathcal{U}_m} \text{rate}_i(t) \leq C_m, \quad (14)$$

where $\mathcal{U}_m$ is the set of users associated with access point m , and C_m is its service capacity.

The mechanism can be directly mapped to existing campus AAA infrastructures through RADIUS Change of Authorization (CoA). In particular, high-reputation users may be granted higher bandwidth caps, elevated scheduling priority, or less restrictive access policies, whereas low-reputation users may be rate-limited, deprioritized, or forced to disconnect and re-authenticate [6]. This engineering mapping provides a practical path for embedding the proposed game-theoretic mechanism into real campus-network control loops.

Given the above design, the system can be analyzed by first solving $\pi_S(x) = \pi_D(x)$ under a homogeneous approximation to identify possible interior fixed points. The convergence and local stability of the mechanism are then evaluated through continuous and discrete replicator dynamics, while the fixation probability and absorption behavior in finite populations are further examined through the Moran process [7].

IV. SIMULATION SETUP AND RESULTS

The campus network is modeled as a three-layer topology consisting of buildings, access points (APs), and users. The building layer is represented by either a grid graph or a random geometric graph, with M APs and N users. Users are associated with APs according to the network topology, and the initial sharing ratio $x(0)$ is specified in the parameter file. The AP load and user-association states are recorded throughout the simulation [5].

To evaluate the stability and applicability of the proposed mechanism, parameter scanning is performed over the reputation threshold θ , reward intensity α , punishment intensity γ , smoothing factor λ , contribution baseline κ , detection probability p_d , and false-punishment probability p_f . Such adaptive strategy evaluation is related to learning-based dynamic spectrum access, where users update access decisions under changing wireless conditions [12]. Unless otherwise stated, the physical-layer rate model, time-step size, and initial reputation r_0 are explicitly specified in the configuration file [4]. For reproducibility, the default random seed is set to 20260318, and R independent runs are performed with run-dependent seeds.

Three baseline settings are considered for comparison: a no-reputation baseline, in which only the basic access process is retained; a fixed-reward baseline, in which sharing users receive a constant reward without punishment; and the proposed reputation-based incentive and punishment mechanism. The main evaluation metrics include the sharing ratio $x(t)$, system throughput $\text{Th}(t)$, Jain's fairness index $J(t)$, and convergence time T_s . The sharing ratio is defined as

$$x(t) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}_{\{a_i(t)=S\}}, \quad (15)$$

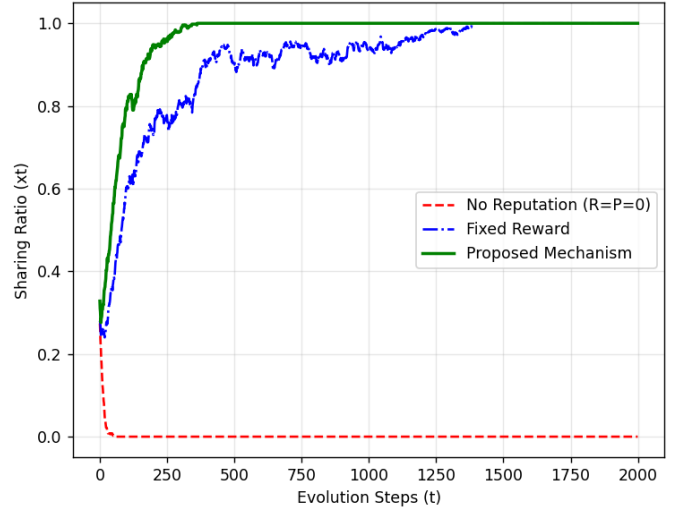


Fig. 1. Evolution of the sharing ratio under different mechanisms. The proposed reputation-based incentive and punishment mechanism converges faster and achieves a higher steady-state sharing level than the no-reputation and fixed-reward baselines.

the throughput is

$$\text{Th}(t) = \sum_i \text{rate}_i(t), \quad (16)$$

and Jain's fairness index is

$$J(t) = \frac{(\sum_i \text{rate}_i(t))^2}{N \sum_i \text{rate}_i^2(t)}. \quad (17)$$

The convergence time is defined by

$$T_s = \min \{t : |x(t) - x(t-1)| < \epsilon_s\}, \quad (18)$$

where ϵ_s is a prescribed tolerance. Jain's fairness index adopts the standard formulation in [10].

Fig. 1 compares the evolution of the sharing ratio under different mechanisms. Without reputation regulation, the sharing ratio rapidly drops to zero, indicating that free-riding becomes dominant. The fixed-reward baseline improves the steady-state sharing level, but its convergence is slower and less stable. In contrast, the proposed mechanism drives the sharing ratio to converge quickly to a near-full-sharing state, showing that the joint reward-punishment design is more effective in sustaining cooperative behavior.

Fig. 2 further shows the final distribution of user reputation scores after 2000 evolution steps. The dashed vertical line indicates the reputation threshold $\theta = 0.35$. Most users remain above the threshold, suggesting that the proposed mechanism forms a relatively stable high-reputation population and effectively suppresses persistent free-riding behavior.

Overall, the results indicate that suitable combinations of θ , α , and γ can improve both the steady-state sharing level and the stability of the system. The recorded outputs include at least the trajectory of the sharing ratio, the terminal reputation distribution, the system throughput, and the user-utility distribution, thereby ensuring that the results are repeatable, comparable, and auditable.

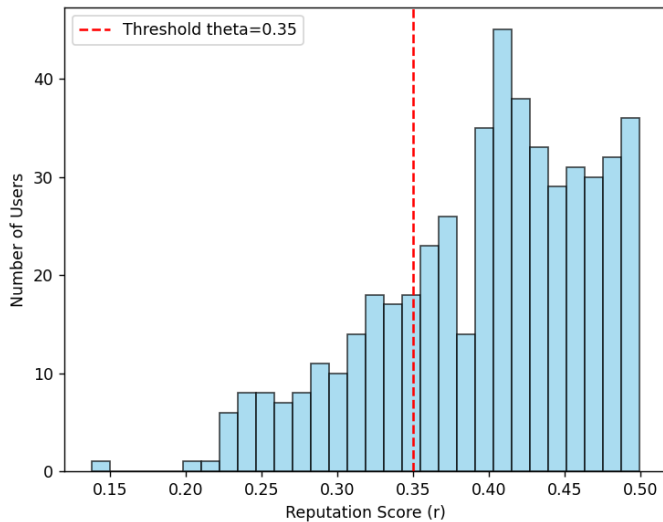


Fig. 2. Final reputation-score distribution after 2000 evolution steps. The dashed vertical line denotes the reputation threshold $\theta = 0.35$. Most users are concentrated above the threshold, indicating that the proposed mechanism promotes a stable high-reputation population.

V. CONCLUSION

This paper addressed the free-rider problem in campus WiFi sharing by developing a unified reputation-threshold incentive and punishment framework under evolutionary game theory. The proposed framework integrates replicator dynamics, discrete-time updating, and the finite-population Moran process to characterize strategic evolution in continuous-population, slot-based, and finite-user settings. In addition, by leveraging the dynamic authorization capability of RADIUS Change of Authorization (CoA), the reputation mechanism can be mapped to executable session-level control actions, which improves the practical deployability of the model.

Several limitations remain in the current study. Some physical-layer parameters, capacity constraints, and rate models have not yet been fully instantiated, and the experimental section still requires more complete numerical results and statistical validation. Future work may proceed in three directions: first, incorporating real or semi-real campus topologies and traffic workloads to improve simulation fidelity; second, extending the model to multi-strategy scenarios involving more complex behaviors such as Sybil attacks, whitewashing, and malicious abuse; and third, replacing the piecewise threshold-based reward and punishment rule with a continuous control function to further enhance fairness and robustness against misclassification [4], [9].

REFERENCES

- [1] *IEEE Standard for Information Technology—Telecommunications and Information Exchange Between Systems Local and Metropolitan Area Networks—Specific Requirements—Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications*, IEEE Std. 802.11-2020, 2020.
- [2] B. Aboba, L. Blunk, J. Vollbrecht, J. Carlson, and H. Levkowitz, “Extensible Authentication Protocol (EAP),” RFC 3748, Jun. 2004.
- [3] A. Rubens, C. Rigney, S. Willens, and W. A. Simpson, “Remote Authentication Dial In User Service (RADIUS),” RFC 2865, Jun. 2000, doi: 10.17487/RFC2865.
- [4] S. J. Wang and Z. Y. Wang, “An evolutionary game model with reputation threshold and reputation score to promote trust in the sharing economy,” *Scientific Reports*, vol. 15, Art. no. 14635, 2025, doi: 10.1038/s41598-025-98728-z.
- [5] Y. Ma, Q. Zhang, F. Li, et al., “IPoE based campus network IPv4/IPv6 dual-stack admission control design and implementation,” *Journal of Shenzhen University Science and Engineering*, vol. 37, suppl. 1, pp. 1–5, 2020, doi: 10.3724/SPJ.1249.2020.99001.
- [6] G. Dommety, M. Chiba, D. Mitton, M. Eklund, and B. Aboba, “Dynamic Authorization Extensions to Remote Authentication Dial In User Service (RADIUS),” RFC 5176, Jan. 2008, doi: 10.17487/RFC5176.
- [7] R. Cressman and Y. Tao, “The replicator equation and other game dynamics,” *Proceedings of the National Academy of Sciences of the United States of America*, vol. 111, suppl. 3, pp. 10810–10817, 2014.
- [8] H. Ohtsuki, P. Bordo, and M. A. Nowak, “The one-third law of evolutionary dynamics,” *Journal of Theoretical Biology*, vol. 249, no. 2, pp. 289–295, Nov. 2007, doi: 10.1016/j.jtbi.2007.07.005.
- [9] C. Wang, M. Perc, and A. Szolnoki, “Evolutionary dynamics of any multiplayer game on regular graphs,” *Nature Communications*, vol. 15, Art. no. 5349, Jun. 2024, doi: 10.1038/s41467-024-49505-5.
- [10] R. Jain, D.-M. Chiu, and W. R. Hawe, “A quantitative measure of fairness and discrimination for resource allocation in shared computer systems,” DEC Research Report TR-301, Sep. 1984.
- [11] H. Hu, L. Li, W. Lin, X. Li, and Z. Han, “Reflection coefficients optimization in IRS-assisted communication: An evolutionary game approach,” in *Proc. 13th International Conference on Wireless Communications and Signal Processing (WCSP)*, Changsha, China, Oct. 2021, pp. 1–5, doi: 10.1109/WCSP52459.2021.9613652.
- [12] P. Yang, L. Li, J. Yin, H. Zhang, W. Liang, W. Chen, and Z. Han, “Dynamic spectrum access in cognitive radio networks using deep reinforcement learning and evolutionary game,” in *Proc. IEEE/CIC International Conference on Communications in China (ICCC)*, Beijing, China, Aug. 2018, pp. 405–409, doi: 10.1109/ICCCChina.2018.8641242.
- [13] Y. Liu, L. Li, W. Liang, H. Zhang, A. Gao, X. Li, and Z. Han, “Auction-based users dynamic allocation in heterogeneous networks,” in *Proc. IEEE/CIC International Conference on Communications in China (ICCC)*, Beijing, China, Aug. 2018, pp. 636–640, doi: 10.1109/ICCCChina.2018.8641221.